

Pattern Recognition 2024: Homework 5

周韧平– zrp21@mails.tsinghua.edu.cn

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Problem 1

(i)

$$\begin{aligned}\mathcal{L}(\theta) &= \log(p(\mathbf{X}|\theta)p(\theta)) \\ &= \sum_{i=1}^n \log\left(\sum_{z_i} p(x_i, z_i; \theta)\right) + \log(p(\theta)) \\ &\geq \sum_{i=1}^n \log\left(\sum_{z_i} q(z_i) \log(p(x_i, z_i|\theta)) - \sum_{z_i} q(z_i) \log(q(z_i))\right) + \log(p(\theta))\end{aligned}$$

推广后的E-step和M-step分别为

E-step:

$$\begin{aligned}q_{[k+1]}(z) &= p(z|x, \theta_{[k]}) \\ F(q_{[k+1]}, \theta) &= \sum_{z_i} q_{[k+1]}(z_i) \log(p(x_i, z_i|\theta)) - \sum_{z_i} q_{[k+1]}(z_i) \log(q_{[k+1]}(z_i)) + \log(p(\theta))\end{aligned}$$

M-step:

$$\theta_{[k+1]} = \operatorname{argmax}_{\theta} Q(\theta_{[k]}, \theta) = \operatorname{argmax}_{\theta} \sum_z p(z|x, \theta_{[k]}) \log(p(x, z|\theta_{[k]})) + \log(p(\theta))$$

由此可见M-step是最大化 $\log(p(x, z|\theta_{[k]}))$ 与 $\log(p(\theta))$ 的线性组合, 因此该步骤可解, 又因为 $L(\theta_{[k+1]}) - L(\theta_{[k]}) = Q(\theta_{[k+1]}, \theta) - Q(\theta_{[k]}, \theta)$, 由于每次都是取满足最大值的 θ , 因此 $L(\theta)$ 单增

(ii) 经过200步迭代后，参数估计的结果为

$$\begin{aligned}\pi &= [0.678251320.32174868] \\ \mu_1 &= [-0.957445980.00701796]^T \\ \mu_2 &= [0.97406856 - 0.0204322]^T \\ \Sigma_1 &= \begin{bmatrix} -0.95744598 & 0.00701796 \\ 0.97406856 & -0.0204322 \end{bmatrix} \\ \Sigma_2 &= \begin{bmatrix} 0.53120883 & 0.01915489 \\ 0.01915489 & 0.54821003 \end{bmatrix}\end{aligned}$$

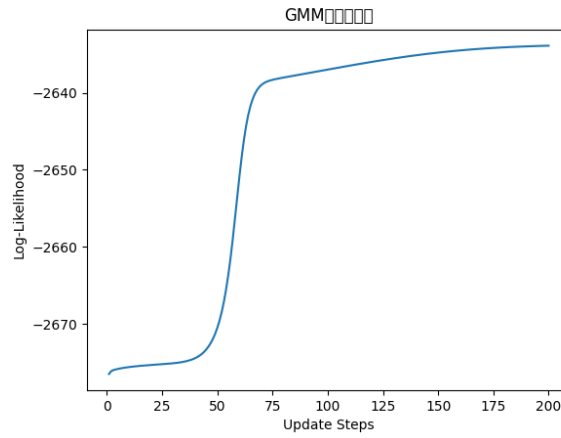


Figure 1: 对数似然结果